

Does Sim-to-Sim Predict Sim-to-Real? Multimodal Diffusion Inverse Kinematics on a Physical Tendon-Driven Continuum Arm

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Abstract—Conditional diffusion models can represent the full, multimodal solution set of a redundant inverse kinematics (IK) problem, and a prior controlled study established three things in simulation: the advantage over deterministic regressors is specifically a redundancy effect, the learned inverse map does not survive a change of dynamics model, and the recovery mechanisms available after that collapse are sharply bounded. That study was simulation-only. We port its full protocol to a physical two-section, six-tendon continuum arm with camera-based shape measurement, and ask two questions. First, does the transfer collapse reproduce against real hardware, where the source model is a calibrated model of the very arm being commanded? Second, and more useful to the field, does the cheap sim-to-sim characterization *rank* methods the same way expensive hardware evaluation does—i.e., can sim-to-sim serve as a screening proxy? We describe the platform, the measurement chain, and a protocol in which every reported gap is bounded below by an independently measured noise floor. **[TODO: one sentence of headline results once measured]**

Index Terms—Modeling, Control, and Learning for Soft Robots; Deep Learning Methods; inverse kinematics; sim-to-real transfer.

I. INTRODUCTION

A tendon-driven continuum arm with three tendons per section has more actuation degrees of freedom than task degrees of freedom: a two-section arm exposes six actuator commands to specify a three-dimensional tip position. The inverse map is therefore one-to-many, and its preimages are not merely a connected null space—physically distinct arm shapes reach the same point. A deterministic learned regressor trained on motor-babbling data [7] fits the conditional *mean* of that multimodal distribution and can return an actuation realizing none of the true solutions, while a conditional diffusion model [2], [3] learns the full conditional distribution and can sample every solution mode.

A recent controlled simulation study [1] established three results about this setting. (i) The diffusion advantage is *specifically* redundancy resolution rather than generic model capacity: removing redundancy by conditioning on whole-body shape collapses sampled diversity by roughly 80× and nearly erases the gap to a deterministic baseline, while adding redundancy by increasing segment count widens it monotonically. (ii) The learned inverse map is simulator-specific: actuations from a model trained on a piecewise-constant-curvature (PCC) simulator collapse when executed on a Cosserat-rod simulator of the same arm, on every training seed. (iii) The recovery available after that collapse is bounded—candidate selection under a target-domain query budget helps but never solves a target, a diversity-

aware query ordering significantly outperforms confidence ordering, guidance-based adaptation is at best suggestive, gain domain randomization achieves nothing, and target-domain fine-tuning buys genuine transfer only across a sharp retention-transfer cliff that data mixing cannot smooth.

Every one of those results was obtained between two simulators. That is a deliberate methodological choice—it isolates model mismatch from sensing noise, actuator imperfection, and unmodeled contact—but it leaves the practically decisive question untouched. This paper addresses it directly.

Why hardware changes the question

Moving to a physical arm does more than add realism; it inverts the relationship between the source model and the target domain. In the prior study, the source simulator (PCC) was never intended as a model of the target simulator (Cosserat rod); the two were independently specified, and their disagreement was structural and directional. Here, by contrast, the source model is a PCC model whose parameters are *fitted to the very arm the policy will command*. If the transfer collapse survives even that best-case pairing, the finding is considerably stronger than its sim-to-sim predecessor. If it does not, the magnitude of the sim-to-sim gap was an artifact of pairing two unrelated models, and that is worth reporting with equal prominence.

Hardware also makes one of that study’s framings natural rather than contrived. The query-budget experiment asked how many *expensive* target-domain evaluations candidate selection requires; in simulation an “expensive” evaluation was four seconds of Cosserat integration. On hardware it is a physical actuation, a settling interval, and a camera measurement—genuinely costly, non-batchable, and subject to wear. The economics the experiment was designed to probe are real here.

Contributions

- 1) **An inexpensive, reproducible hardware platform and protocol.** A two-section, six-tendon continuum arm built from commodity parts (**[TODO: final bill-of-materials cost]**), with camera-based shape measurement, and a full port of that study’s evaluation suite. All fabrication templates, calibration procedures, and code are released.
- 2) **A sim-to-real characterization of multimodal diffusion IK.** We measure whether the transfer collapse, the redundancy-specific advantage, and the bounded-recovery result each reproduce against physical hardware (Sec. VI).

- 3) **A cross-domain consistency test.** We compare the *ranking* of methods and recovery mechanisms under the prior sim-to-sim evaluation against their ranking on hardware. Agreement would license sim-to-sim screening as a cheap proxy for hardware evaluation; disagreement would caution against a practice the field currently relies on implicitly.

Because a physical platform introduces error sources that simulation does not, we treat measurement rigour as a first-class part of the method rather than an appendix. Sec. IV-C specifies an independently measured noise floor, and no reported gap is interpreted unless it exceeds that floor.

II. RELATED WORK

Continuum-arm IK. Non-learned approaches invert a model at query time—Jacobian schemes for nonconstant-curvature arms [7] and real-time FEM inversion [9]. The constant-curvature kinematics underlying our source model are reviewed in [8], and tendon-driven construction with spacer discs follows standard practice in the continuum-robotics literature. Learned approaches fit a deterministic target-to-actuation map on babbling data [7]; mixture density networks [10] are the classical multimodal alternative.

Diffusion models for robot action. Denoising diffusion models [2] with DDIM sampling [3] and classifier-free guidance [4] underpin diffusion policies [5]. IKDiffuser [6] solves IK for rigid multi-arm systems with flexible kinematic-tree topology; compliant continuum bodies are outside its scope.

Crossing the reality gap. Domain randomization [11] and target-domain fine-tuning are the standard levers, with catastrophic forgetting under naive fine-tuning well-documented [12]. What is comparatively rare, and what this paper supplies, is a single method evaluated under *both* a controlled sim-to-sim gap and a physical sim-to-real gap with a shared protocol, permitting the two to be compared rather than merely reported side by side.

III. HARDWARE PLATFORM

A. Mechanical design

The arm is a two-section tendon-driven continuum manipulator built around a single flexible backbone. The backbone is a 3.18 mm (1/8") structural fiberglass rod of free length 305 mm (12"). Six spacer discs of 56 mm diameter and 3.0 mm thickness are bonded to the backbone at a uniform 50.8 mm (2") pitch with two-part epoxy, dividing the arm into two actuated sections of three discs each.

Backbone diameter is set by a force budget rather than by convenience, and it dominates every other mechanical choice. Bending a section to curvature κ requires a moment $EI\kappa$, supplied by tendon tension acting at the outer tendon radius, and I grows as the fourth power of diameter. At the 22 mm moment arm used here, one servo at half stall bends a 1/8" rod roughly 90–150° per section but a 3/16" rod only 20–30°: a $5.1\times$ stiffness difference that shrinks the reachable workspace by a factor of 50, to a 30 mm-deep shell in which the tip never leaves full extension. Distinct solutions remain resolvable on the stiffer rod—they separate

by ≈ 11 mm against a millimetre-scale sensing floor—so the choice is not between a working and a broken platform, but the redundancy the study exists to characterize is largely absent from the stiffer one.

Discs are fused-deposition printed in PLA rather than cut and drilled, for three reasons that bear on measurement quality. The central bore is a slip fit on the backbone, so each disc is held square by its own bore; a drilled clearance hole in stock disc material permits several degrees of tilt before the epoxy sets. The marker tab is part of the disc, so its 65° presentation angle is identical on all six by construction rather than by hand-gluing—and tab orientation is the single variable simultaneous marker visibility is most sensitive to. Finally, hole positions are reproduced from one geometric description, so disc-to-disc scatter is limited by printer repeatability rather than by jig wear.

Each disc carries seven holes: a central clearance hole for the backbone and six tendon guides arranged on two concentric circles at radii 12.0 mm and 22.0 mm, with three holes per circle at 90°, 210° and 330°. An index notch on the rim marks the 90° heading so that all six discs are keyed to a common angular reference during assembly. Section A's three tendons run on the inner circle and terminate at disc 3; section B's run on the outer circle, pass freely through discs 1–5, and terminate at disc 6. Both sections therefore actuate along *identical* angular headings and differ only in moment arm—a deliberate choice that preserves the three-fold symmetry assumed by the constant-curvature source model, so that the two sections are related by a single scalar per section rather than by a change of basis.

Tendons are 0.79 mm (1/32") 7×7 stainless wire rope, terminated with swaged sleeves and pre-tensioned through inline turnbuckles at the base. Steel rope is used rather than braided polymer line specifically to suppress creep: commanded tendon length must remain a stable function of actuator angle across thousands of cycles.

Because all six discs are printed from one geometric description, per-disc random error—which no calibration can absorb—is replaced by a single systematic bias common to every disc, which one gain fit per section absorbs exactly. Printed holes come out undersized by a machine-dependent amount, so the bore is sized from a printed gauge coupon rather than nominally, and the same offset applies to all six. **[TODO: report measured disc-to-disc hole scatter and the bore offset used]**

B. Actuation

Six ROBOTIS DYNAMIXEL XL330-M288-T smart servos, one per tendon, are mounted on a rigid 305 $\times$ 305 mm acrylic base plate in three pairs at 120° about the backbone, aligning each servo pair with the tendon heading it drives. Each servo carries a metal horn with a brass standoff acting as a cable capstan; tendon displacement is related to commanded servo angle by $\Delta\ell = \theta(r_s + r_c)$ with r_s the capstan radius and r_c the cable radius, measured empirically rather than assumed (Sec. V-A). Cable is wound in a single layer so that the effective radius, and hence this relation,

remains constant over the working range. Servos operate in extended-position (multi-turn) mode with software limits set below the tendon’s mechanical travel.

The base plate is drilled with the same seven-hole pattern as a spacer disc and functions as disc V_0 : tendons emerge from the plate on the same radii and headings they occupy at every disc above, so that no tendon runs obliquely between the base and the first disc.

C. Command integrity

One implementation detail is reported because it is invisible until it corrupts data. The six servos share a single half-duplex bus and are commanded with a synchronised broadcast write, so that both sections actuate simultaneously rather than in sequence. That broadcast returns no acknowledgement by design; the host library therefore reports success as soon as the bytes are queued, whether or not any servo acted on them. If the next packet is placed on the bus too soon, the servos discard the write *silently*—no error code, no fault flag, and every register still reading back the expected configuration.

We characterised the threshold directly by varying the interval between the broadcast write and the following read, and counting how many servos executed the command: none at 0 or 5 ms, and all six at 10 ms and above. All experiments enforce a 25 ms interval, roughly $2.5\times$ the measured threshold. Without it the failure does not announce itself—it appears in the data as an actuation that was commanded but never applied, which is indistinguishable from a genuine tracking failure and would inflate any reported sim-to-real gap. We note it because anyone reproducing this platform on a shared servo bus will meet it.

D. Actuation parameterization

To keep the interface identical to the prior study, the policy commands a normalized actuation vector $\mathbf{q} \in [0, 1]^6$, mapped affinely onto each servo’s calibrated travel range. Per-section curvature components follow the same three-chamber form used for the pneumatic arm in [1], with tendon displacements substituted for chamber pressures:

$$k_x = g(\ell_0 - \tfrac{1}{2}\ell_1 - \tfrac{1}{2}\ell_2), \quad (1)$$

$$k_y = g \tfrac{\sqrt{3}}{2}(\ell_1 - \ell_2), \quad (2)$$

with g a per-section gain fitted in Sec. V-A. This substitution is what allows the entire software stack, the trained policies, and the evaluation suite to transfer unmodified.

IV. MEASUREMENT SYSTEM

A. Vision chain

Arm shape is measured by a fixed monocular global-shutter camera (OV9281, 1280×800 , monochrome) observing planar fiducial markers mounted flat on the spacer discs. A global shutter is used rather than a rolling shutter so that measurements taken during residual motion are not geometrically distorted—relevant because, as in the prior study’s target domain, the arm is sampled at a fixed settling horizon rather than at true static equilibrium.

Markers are ArUco fiducials [13] of side 30 mm, the largest that fits the 40 mm printed marker tab with an adequate quiet zone. Marker side is measured with calipers rather than taken as nominal: monocular pose recovers range from apparent size, so a marker declared 5% large biases every reported position 5% far—on this arm roughly 15 mm of systematic error that averaging cannot remove and that presents exactly as a transfer gap. Markers on discs 3 and 6 give section-boundary and tip pose; a full set on all six discs gives whole-backbone shape for the shape-conditioning experiment. Detection and pose estimation use OpenCV [14].

Camera intrinsics and distortion are estimated once from [TODO: N] checkerboard views and held fixed. The camera-to-base extrinsic transform is established from a marker at a surveyed position on the base plate, and the camera is not moved thereafter; every reported position is expressed in the base frame.

B. Sampling protocol

One sample comprises: draw an actuation, command all six servos, wait a fixed settling interval of [TODO: **settle time**] s, capture one frame, detect markers, and record the actuation together with the resulting disc poses. Detections failing a confidence or visibility check are discarded and redrawn; we report the discard rate as a data-quality statistic, a diagnostic with no analogue in simulation.

The settling interval is fixed rather than adaptive, making the target domain a deterministic dynamic-snapshot protocol—the same convention adopted, and explicitly flagged, in that study’s Cosserat domain. It is applied identically to every sample, every dataset, and every evaluation here.

C. Noise floor and repeatability

Because the central claim concerns the size of a measured gap, we bound below what a gap can mean. Two quantities are measured before any policy is evaluated.

Sensing noise. Two markers are fixed at a separation measured independently with digital calipers, and the vision chain estimates that separation [TODO: N] times under the operating configuration. The spread of those estimates is the sensing noise floor. [TODO: **report mean error and standard deviation**]

Actuation repeatability. A set of [TODO: N] actuations is commanded [TODO: k] times each, in randomized order with a full return to the home configuration between repeats. The dispersion of measured tip position for a repeated command is the actuation repeatability floor, and it folds in servo backlash, cable hysteresis, and any residual creep. [TODO: **report repeatability**]

A reported transfer gap is interpreted only where it exceeds the combination of these two floors. This is the substitute, on hardware, for the exact determinism a simulator provides for free.

V. SOURCE MODEL, TRAINING, AND PROTOCOL

A. Fitted source model

The source domain is a differentiable PCC model of *this* arm, with parameters fitted to it rather than assumed. Calibration proceeds in two stages. First, each capstan’s effective radius is measured directly by commanding a known servo rotation and measuring paid-out cable with calipers, yielding the angle-to-tendon-length relation. Second, the per-section curvature gain g is fitted by least squares over [TODO: N] calibration actuations, minimizing measured-versus-predicted disc pose. [TODO: report fitted gains and calibration residual]

This is the pairing that makes the hardware experiment sharp. The source model is not an arbitrary alternative dynamics—it is the best constant-curvature description of the physical arm that the data supports, and its residual is reported so readers can judge how good a source model the policy was given.

B. Fit quality as a reported quantity

Because the source model is fitted rather than assumed, its residual is a result, not a footnote: it upper-bounds how much of any observed transfer gap is attributable to an imperfect source model rather than to a genuine limitation of the learned inverse map. We report the distribution of measured-versus-predicted disc position error, and compare it against the sensing noise floor of Sec. IV-C. A residual close to that floor means the constant-curvature description fits this arm about as well as the instrumentation can resolve, and the transfer gap may be read at face value; a residual far above it bounds how much of the gap is our own modelling error. [TODO: report fitted gains and residual distribution]

C. Training data and models

Training data are [TODO: N] actuations drawn i.i.d. uniformly and pushed through the fitted PCC model. We evaluate the same four methods as the prior study, with architectures and hyperparameters carried over unchanged so that any difference in outcome is attributable to the domain rather than to retuning: a conditional diffusion model (residual FiLM-conditioned MLP denoiser, DDPM training with cosine schedule and classifier-free guidance, DDIM sampling), a deterministic MLP regressor, a mixture density network [10], and a random-restart gradient-IK solver optimizing through the differentiable fitted model. Unless stated otherwise K candidates are sampled per target and best-of- K is reported; success is tip error below [TODO: tolerance] mm, set to the same fraction of arm length used previously.

D. Experiments

- **H1 (source-domain accuracy).** Tip error against the fitted PCC model, verifying the policies are trained correctly before hardware is involved. This isolates training failures from transfer failures.
- **H2 (multimodality on hardware).** Ground-truth solution sets are enumerated on the fitted source model,

and mode recall is scored using *measured* configurations. Distances are computed in curvature space rather than actuation space, since the tendon-to-curvature map carries a per-section null space that would otherwise contaminate distance-based metrics.

- **H3 (downstream value).** Obstacle avoidance with physical obstacles at surveyed positions; success requires some candidate to be both accurate and collision-free.
- **H4 (sim-to-real transfer).** The headline experiment: actuations from the source-trained policies executed on the arm, measured by camera, and compared against source-domain performance.
- **H5 (recovery mechanisms).** Candidate selection under a physical query budget, diversity-aware versus confidence query ordering, and fine-tuning on measured data across a range of source-data mixing ratios—testing in particular whether the retention-transfer cliff persists when the target-domain data are real measurements rather than simulator output.

E. Cross-domain comparison

The consistency test of Contribution 3 compares method and mechanism *rankings* between the prior sim-to-sim study and H1–H5 here. We deliberately do not fit a relationship between the two gap magnitudes: with a single hardware platform, that would be an $n=1$ regression. Rank agreement is what a practitioner would actually rely on when using a cheap sim-to-sim study to decide which method to carry forward to hardware, and rank agreement is what we report.

VI. RESULTS

[TODO: Entire section. Populate from H1–H5. Suggested structure: Table I comparing all four methods in source domain and on hardware; Fig. 2 query budget curve with physical queries; Table II fine-tuning mixing grid; Fig. 3 rank comparison against the sim-to-sim study.]

VII. DISCUSSION AND LIMITATIONS

[TODO: Entire section. Limitations to state explicitly once results exist: single hardware platform ($n=1$ for the cross-domain claim); planar or near-planar sensing if a single camera is used; open-loop IK formulation with best-of- K presupposing a verifier; fitted PCC as source model is a best-case pairing and a harsher source model would likely widen the gap.]

REPRODUCIBILITY

Code, fabrication templates, calibration procedures, and per-experiment result files are available in an anonymized repository: [TODO: anonymous.4open.science mirror URL - mint it from a fresh branch].

REFERENCES

- [1] **[TODO: The sim-to-sim paper. Under double-anonymous review cite it in the third person as an existing preprint - never as "our previous work", and do not include a repository URL that identifies the authors.]**
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